

Question 1

What is a variable in C/C++?

A variable is a named memory location used to store data, and its value can be changed during program execution.

Question 2

- Why do we use variables in programs?

We use variables in a program to store data so that it can be accessed, modified, and used during program execution.

Question 3

- What is the difference between a variable name and its value?

A variable name is the identifier used to refer to a memory location, whereas the value is the data stored in that memory location.

Question 4

- Which of the following is a valid variable name? a) 2num b) total_marks c) for d) my marks

Ans-b) total_marks

Question 5

- Why is Age different from age?
- Ans- Age and age are different because C and C++ are case-sensitive languages. Uppercase and lowercase letters are treated as different characters.

Question 6

- Can a variable name contain spaces? Why?
- Ans - No, a variable name cannot contain spaces because the compiler treats the space as a separator between two different tokens (identifiers or keywords), which results in a syntax error.

Question 7

- Can a variable name start with an underscore (`_`) ?
- Ans- Yes, a variable name can start with an underscore (`_`), but it should be avoided because such names may be reserved for the compiler or system.

Question 8

- What happens if two variables have the same name in the same scope?
- Ans - If two variables have the same name in the same scope, the compiler generates a redeclaration error because two variables cannot have the same identifier in the same scope.

Question 9

- Write three meaningful variable names for storing student information.
- studentname, rollnumber, studentage

Question 10

- Which naming convention is better and why: studentMarks or sm?
- Ans - studentMarks is a better variable name because it is meaningful, easy to understand, and improves the readability and maintainability of the code.

Question 11

- What is a data type?

Ans -A data type specifies the type of data that a variable can store in a program

Ex-int ,float, char,double

Question 12

- Which data type is used to store whole numbers?
- Ans - int

Question 13

- Which data type is used to store decimal values?
- Ans- float and double

Question 14

- What is the difference between float and double?
- Ans - float is a single-precision floating-point data type that typically uses 4 bytes of memory, while double is a double-precision floating-point data type that typically uses 8 bytes. double provides higher precision and can store more decimal digits than float.

Question 15

- Which data type stores a single character?
- Ans - char is the data type used to store a single character.

Question 16

- Which data type stores True/False values?
- Ans – bool is the data type used to store true or false

Question 17

- What data type is suitable for storing a person's name?
- Ans – string

Question 18

- Predict the output: `int age=18; cout<<age;`
- Ans - 18

Question 19

- Identify the data type: 25, 3.14, 'A', true.
- Ans - 25=Int,3.14=float,A=char,true=boolean

Question 20

- What happens if you store a decimal number in an int variable?
- Ans -When a decimal value is assigned to an int variable, the decimal part is truncated, and only the whole number is stored.
- Example – 4.56 will become 4

Question 21

- What is the purpose of cin?
- Ans - cin is a predefined input stream object in C++ used to take input from the user through the keyboard.

Question 22

- What is the purpose of cout?
- Ans -cout is a predefined output stream object in C++ used to display (print) output on the screen

Question 23

- Which symbol is used with cin?
- Ans -cin uses the >> (extraction operator) to take input from the user.

Question 24

- Which symbol is used with cout?
- Ans -cout uses the << (insertion operator) to display output on the screen

Question 25

- Explain: `cin >> age;`
- Ans -`cin >> age;` takes an integer value as input from the user and stores it in the variable `age`

Question 26

- Explain: `cout << age;`
- Ans -`cout << age;` displays the value stored in the variable `age` on the screen.

Question 27

- What is the purpose of endl?
- Ans - endl is used to move the cursor to the next line and flush the output stream.

Question 28

- Write a statement to input a student's marks.
- Ans -int marks;
- cin >> marks;

Question 29

- Write a statement to display 'Welcome to C++'.
- `Cout<<"welcome to c++";`

Question 30

- Predict the output: `cout << 'Programming';`
- Programming

Question 31

- What is an operator?
- Ans - An operator is a symbol that performs an operation on one or more operands (values or variables).

Question 32

- Which operator is used for addition?

Ans - The + (plus) operator is used for addition

```
int a = 10, b = 20;
```

```
int sum = a + b;
```


Question 33

- What is the result of $10 \% 3$?

Ans - The result of $10 \% 3$ is - **1**

because $\%$ (modulus) operator returns the remainder after division

Question 34

- What is the result of $8 \% 2$?

Ans-

The result of $8 \% 2$ is 0.

Question 35

- Which operator checks equality?

Ans –

The == operator checks whether two values are equal.

Question 36

- What is the difference between = and == ?

Ans –

= is the assignment operator .It assigns a value to a variable

== is the equality operator.It compares two values and returns true if they are equal,otherwise false.

Question 37

- What is the result of $5 > 3$?

Ans – Result of $5 > 3$ is true ,because 5 is greater than 3

Question 38

- What does && represent?

Ans –

&& is the logical AND operator. It returns true only if both conditions are true; otherwise, it returns false.

Question 39

- What does `||` represent?

Ans –

`||` is the logical OR operator. It returns true if at least one of the conditions is true; otherwise, it returns false.

Question 40

- What does ! represent?

Ans –

! generally represents the logical NOT (negation) operator. It reverses the truth value of a condition or boolean expression (true becomes false, and false becomes true).

Question 41

- Write a program to print 'Hello World'.

Ans –

```
#include <iostream>
using namespace std;
```

```
int main()
{
    cout << "Hello, World!" << endl;

    return 0;
}
```

Question 42

- Write a program to input two numbers and display their sum.

Ans – #include <iostream>

using namespace std;

```
int main()
{
    int num1, num2, sum;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    sum = num1 + num2;

    cout << "Sum = " << sum << endl;

    return 0;
}
```

Question 43

- Write a program to calculate the area of a rectangle.
- Ans - #include <iostream>
- using namespace std;
- int main()
- {
- float length, breadth, area;
- cout << "Enter length: ";
- cin >> length;
- cout << "Enter breadth: ";
- cin >> breadth;
- area = length * breadth;
- cout << "Area of Rectangle = " << area << endl;
- return 0;
- }

Question 44

- What formula is used to calculate Simple Interest?
- Ans - $\text{simpleInterest} = (\text{principal} * \text{rate} * \text{time}) / 100.$

Question 45

- Write a program to calculate Simple Interest.
- Ans - `#include <iostream>`
- `using namespace std;`
- `int main()`
- `{`
- `float principal, rate, time, simpleInterest;`
- `cout << "Enter Principal Amount: ";`
- `cin >> principal;`
- `cout << "Enter Rate of Interest: ";`
- `cin >> rate;`
- `cout << "Enter Time (in years): ";`
- `cin >> time;`
- `simpleInterest = (principal * rate * time) / 100;`
- `cout << "Simple Interest = " << simpleInterest << endl;`
- `return 0;`
- `}`

Question 46

- Why is a temporary variable used while swapping two numbers?
- Ans – A temporary variable is used while swapping two numbers to store the value of one variable so it is not lost during the exchange
- EX-int a = 10, b = 20, temp;
- temp = a;
- a = b;
- b = temp;

Question 47

- Swap the values: a=10, b=20.
- Ans –

```
int a = 10, b = 20;
```

```
a = a + b;
```

```
b = a - b;
```

```
a = a - b;
```

Question 48

- What formula is used to convert Celsius into Fahrenheit?
- Ans - $F = (C \times 9/5) + 32$

Question 49

- What is the Fahrenheit value of 0°C ?
- Ans - The Fahrenheit value of 0°C is 32°F .

Question 50

- Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?
- Ans – Integer division can give incorrect results because it discards the decimal part. If both operands are integers, the result is an integer, which can reduce the accuracy of the calculation.